

Task 1

Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)

Identified Normal Form Violation: The Booking_receipt table violates the First Normal Form (1NF).

Explanation: 1NF requires every column in a table to hold atomic (indivisible) values. The seat_numbers column stores a comma-separated list of values (e.g., "12A, 12B") when a booking covers multiple passengers on the same reservation, which makes it a non-atomic (multi-valued) attribute.

Task 2: Step-by-Step Normalization (1NF → 2NF → 3NF)

1NF (First Normal Form) To convert the dataset to 1NF, multi-valued attributes are expanded into separate rows so that each column contains only atomic values.

Booking_receipt_1NF (booking_id, seat_number, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

Primary Key: (booking_id, seat_number)

Foreign Keys: None

2NF (Second Normal Form) To reach 2NF, partial functional dependencies (where non-key attributes depend on only part of the composite primary key (booking_id, seat_number)) are eliminated by splitting the table.

Booking (booking_id, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name)

Primary Key: booking_id

Foreign Keys: None

Passenger_Ticket (booking_id, seat_number, passenger_full_name, passenger_passport_number, ticket_price)

Primary Key: (booking_id, seat_number)

Foreign Key: booking_id REFERENCES Booking(booking_id)

3NF (Third Normal Form) To achieve 3NF, transitive functional dependencies (where a non-key attribute determines another non-key attribute) are removed by creating individual entities.

Passenger (passenger_passport_number, passenger_full_name)

Primary Key: passenger_passport_number

Airport (airport_name, city)

Primary Key: airport_name

Flight (flight_number, airline_name, departure_airport_name, arrival_airport_name) Primary Key: flight_number Foreign Keys:

departure_airport_name REFERENCES Airport(airport_name) arrival_airport_name REFERENCES Airport(airport_name)

Booking (booking_id, flight_number)

Primary Key: booking_id

Foreign Key: flight_number REFERENCES Flight(flight_number)

Ticket (booking_id, seat_number, passenger_passport_number, ticket_price)

Primary Key: (booking_id, seat_number)

Foreign Keys:

booking_id REFERENCES Booking(booking_id)

passenger_passport_number REFERENCES Passenger(passenger_passport_number)

Task 3: (FDs)

The following functional dependencies justify each normalization step:

1. (booking_id, seat_number) -> passenger_passport_number, ticket_price
2. booking_id -> flight_number
3. passenger_passport_number -> passenger_full_name
4. flight_number -> departure_airport_name, arrival_airport_name, airline_name
5. departure_airport_name -> departure_city
6. arrival_airport_name -> arrival_city

Task 4: Evaluating the Flights Table for 3NF

Current Structure: Flights(flight_id, sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at)

Primary Key: flight_id

Functional Dependencies: flight_id -> sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at

3NF Evaluation & Explanation: The Flights table is in 3NF (Third Normal Form):

1. 1NF: All attributes contain atomic values.
2. 2NF: The primary key (flight_id) consists of a single attribute, eliminating any partial dependencies.
3. 3NF: Foreign key attributes such as departing_airport_id, arriving_airport_id, and airline_id do not functionally determine any non-key attributes within this table. Thus, no transitive dependency exists between non-key attributes. Every non-key attribute is directly and fully dependent on the primary key flight_id.

Task 5:

Results

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Query #1

Execution time: 0.19ms

passenger_passport_number	passenger_full_name
N12345678	John Doe
N87654321	Jane Smith
N11223344	Alice Brown
N99887766	Bob Johnson
N55443322	Charlie White

Query #2

Execution time: 0.17ms

airport_name	city
Astana International	Astana
Almaty Airport	Almaty
Heathrow	London
JFK International	New York
Dubai International	Dubai

Results

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Query #3

Execution time: 0.15ms

flight_number	airline_name	departure_airport_name	arrival_airport_name
KC871	Air Astana	Astana International	Almaty Airport
KC951	Air Astana	Almaty Airport	Dubai International
BA123	British Airways	Heathrow	JFK International
DV702	Scat	Astana International	Almaty Airport
EK322	Emirates	Dubai International	Heathrow

Query #4

Execution time: 0.14ms

booking_id	flight_number
101	KC871
102	KC951
103	BA123
104	DV702
105	EK322

Query #5

Execution time: 0.15ms

booking_id	seat_number	passenger_passport_number	ticket_price
101	12A	N12345678	150.00
101	12B	N87654321	150.00
102	04C	N11223344	350.00
103	18F	N99887766	600.00
104	02A	N55443322	120.00

DDL

Database: flightcode:1001

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Results

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Query #1

Execution time: 0.44ms

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
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There are no results to be displayed.

Query #2

Execution time: 0.43ms

airport_id	airport_name	country	state	city	created_at	updated_at
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There are no results to be displayed.

Query #3

Execution time: 0.28ms

passenger_id	first_name	last_name	date_of_birth	gender	country_of_citizenship	country_of_residence	passport_number	created_at	updated_at
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There are no results to be displayed.

Query #4

Execution time: 0.29ms

flight_id	sch_departure_time	sch_arrival_time	departing_airport_id	arriving_airport_id	departing_gate	arriving_gate	airline_id	act_departure_time	act_arrival_time	created_at	updated_at
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There are no results to be displayed.

Query #5

Execution time: 0.31ms

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
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There are no results to be displayed.

Database: flightcode:1001

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Results

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Query #1

Execution time: 0.26ms

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KC	Air Astana	Kazakhstan	2026-09-21 18:14:59.168267	2026-09-21 18:14:59.168267
2	DV	Scat Airlines	Kazakhstan	2026-09-21 18:14:59.168267	2026-09-21 18:14:59.168267

Query #2

Execution time: 0.18ms

airport_id	airport_name	country	state	city	created_at	updated_at
1	Astana International	Kazakhstan	Astana	Astana	2026-09-21 18:14:59.168808	2026-09-21 18:14:59.168808
2	Almaty Airport	Kazakhstan	Almaty	Almaty	2026-09-21 18:14:59.168808	2026-09-21 18:14:59.168808

Query #3

Execution time: 0.23ms

passenger_id	first_name	last_name	date_of_birth	gender	country_of_citizenship	country_of_residence	passport_number	created_at	updated_at
1	Aknur	Yernurkyzy	2008-04-20	Female	Kazakhstan	Kazakhstan	N12345678	2026-09-21 18:14:59.169253	2026-09-21 18:14:59.169253
2	John	Doe	2000-01-01	Male	USA	USA	N87654321	2026-09-21 18:14:59.169253	2026-09-21 18:14:59.169253

Query #4

Execution time: 0.26ms

flight_id	sch_departure_time	sch_arrival_time	departing_airport_id	arriving_airport_id	departing_gate	arriving_gate	airline_id	act_departure_time	act_arrival_time	created_at	updated_at
1001	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712	1	2	A1	B2	1	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712
1002	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712	2	1	C3	D4	2	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712	2026-09-21 18:14:59.169712

Query #5

Execution time: 0.24ms

booking_id	flight_id	passenger_id	booking_platform	created_at	updated_at	status	ticket_price
501	1001	1	Web	2026-09-21 18:14:59.171905	2026-09-21 18:14:59.171905	Confirmed	150.00
502	1002	2	Mobile App	2026-09-21 18:14:59.171905	2026-09-21 18:14:59.171905	Confirmed	120.00